

Research on PEMFC cathode circulation under low-load conditions and its optimal control in FCV power system for long-term durability

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ABSTRACT

In this paper, the mechanism of proton exchange membrane fuel cell (PEMFC) cathode cycling is discussed and the high potential suppression control problem for fuel cell vehicles is investigated by using cathode cycling. Firstly, a rigorous single-cell experimental investigation is performed to reveal the impact mechanism of cathode recirculation on the internal polarization processes of PEMFC. Results show that an increase in the recirculation ratio leads to an increase in the polarization impedance in the low and medium frequency regions but decreases the ohmic impedance. Consequently, when the recirculation ratio increases to 1, the voltage drop rates are 2.2%, 3%, 4.6%, 5.8%, and 7% as the current density varies from 0 to 0.2 A/cm², respectively, and the ohmic impedance decreases by 28.1%. On the basis, the optimal control scheme for high potential suppression based on cathode circulation in vehicular PEMFC system is proposed. The performance comparison of different control method is carried out on a software-in-the-loop testing platform. The results show that, for the same control objective, the model predictive control (T-S MPC) method is 55.5% and 16.7% lower than the proportional integral (PI) controller and the Takagi-Sugeno linear quadratic Gaussian (T-S LQG) in terms of controller input cost. On the contrary, the cumulative error of voltage tracking is reduced by 62% and 30.4% and the voltage response against disturbances is improved by 9.7% and 42.8%. Overall, the T-S MPC achieves the optimal control effect with less control effort.

1. Introduction

With the growing emphasis on carbon emission reduction, the widespread adoption of proton exchange membrane fuel cells (PEMFCs) in automotive applications has shown significant advancement in recent decades. This is attributed to their notable attributes, including high efficiency, zero emissions, low noise, and rapid cold start capabilities [1]. Nevertheless, the distinctive demands of in-vehicle application scenarios necessitate fuel cells to operate under varying conditions such as idling, low load, high load, and variable load. Consequently, the issue of durability emerges as a key challenge [2,3].

1.1. Control technology for long-term durability

To achieve prolonged endurance, such as 5000 h or more, in the operation of vehicle fuel cell systems, efficient controllers must be

designed for proton exchange membrane fuel cell (PEMFC) systems to ensure a stable output of critical control variables [4]. A typical vehicle fuel cell system comprises four auxiliary subsystems integrated with a stack. These auxiliary subsystems collaborate to provide the necessary conditions for the electrochemical reaction in the stack. Consequently, the control challenge in a typical fuel cell system encompasses four components: air supply control, hydrogen supply control, thermal and humidity control, and electrical control [5]. Common control variables include air flow/pressure, hydrogen pressure, temperature, relative humidity, and voltage and power control. The corresponding manipulated variables involve the air compressor speed/backpressure valve position, the position (or number) of the hydrogen proportional valve (or injection valve train), the pump/fan speed, and the humidifier output power. In some advanced PEMFC systems, especially those with a self-humidification function replacing the external humidifier, humidity is transformed from the traditional power control mode to full humidification mode.

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For various control challenges in fuel cell systems, researchers have extensively explored different control algorithms and models. The development of a control-oriented dynamic model serves as a fundamental step for controller design [6,7], especially since advanced control algorithms are often constrained by the availability of commercially accessible hardware. Dynamic models fall into two categories: first-principle mathematical models and data-driven black-box or gray-box models. The commonly used models are 0D or 1D mathematical models built on first principles and simulated using MATLAB/Simulink [8]. These models offer comprehensive insights into the internal state of the system, facilitating the design of observer-based control and estimation techniques—an area of current significance. Additionally, they strike a balance between the high speed and fidelity required for control design. Dynamical models of PEMFC systems, such as the one developed in [9–11] and its variations, have been extensively examined from a control perspective. Numerous fuel cell controllers have been developed based on these dynamic models to address challenges related to strong system nonlinearity, parameter uncertainty, and load disturbance rejection.

Much of the control research is centered around the gas supply system, given that oxygen deficiency significantly impacts fuel cell durability [12]. Initial research on gas supply control aimed to achieve optimal responsiveness and accuracy of the oxygen excess ratio (OER). Proportional integral differential (PID) controllers, renowned for their simplicity and widespread engineering applications, were among the pioneering solutions [11]. An enhanced feed-forward multi-gain scheduling PI controller has been proposed and experimentally validated to enhance performance [13]. To address nonlinearities, fuzzy-PID was introduced to manage the OER with positive outcomes, albeit at the expense of real-time performance with more rules [14,15]. Sliding mode control, a variable structure control method, has demonstrated effective control outcomes in [16–18] due to its advantages of simple design, low computational effort, and robust nonlinearity. As control requirements evolve to handle nonlinearities, disturbances, and parameter uncertainties, advanced model-based control algorithms are gaining favor. A model reference adaptive controller is presented in [19] to enhance system dynamic error. Model predictive control has been explored to optimize anti-disturbance responsiveness and OER accuracy [20–22]. Observer-based algorithms such as sliding mode observer, Kalman filter, and Luenberger observer enhance system lifetime by monitoring internal states [23–25]. Observer-based optimal control methods, aiming to derive optimal control laws by solving cost functions, are proposed to improve OER control effects [26]. Optimal control typically involves offline feedback gains and is computationally less intensive. The active disturbance rejection algorithm based on an extended state observer (ESO) [27] considers nonlinearity, uncertainty, and disturbance as lumped terms, estimated in real time by ESO, effectively eliminating overshoot. Furthermore, some neural network-based control algorithms, while demanding substantial experimental data and computational resources, hold promise from a theoretical and technical development standpoint [28–30].

In addition to the breathing variables, other crucial control variables for fuel cells, concerning efficiency and durability, encompass temperature [31], humidity [32], hydrogen pressure [33], and output voltage [34]. Various controllers have been devised for these parameters, paralleling the well-explored air supply controllers. Examples include PI/PID controllers [32], fuzzy control systems [35], optimal control strategies [36,37], active disturbance rejection control approaches [38], and robust control methodologies [39]. Despite the diverse nature of these control algorithms, they share a common motivation and adhere to a similar control workflow. The distinctions lie in the differential controlled variables, causing variations in manipulated variables across the plant, and the diverse inputs and outputs, leading to distinct model characteristics, such as nonlinearity.

1.2. Cathode recirculation

The control variables strongly affecting fuel cell durability have been summarized from a control perspective, but this is for the conventional vehicle fuel cell system. While there is a wealth of research aimed at enhancing durability from a control perspective, there still exists a notable gap, particularly concerning the issue of carbon corrosion induced by high potentials under low-load conditions [40]. Cathode recirculation is effective in mitigating high potential through the reduction of oxygen concentration in incoming air, thereby alleviating durability degradation caused by carbon corrosion [41]. Numerous studies have been conducted in this area in recent years. In the study by Jiang et al. [42], the performance of cathode recirculation in a 10 kW stack is investigated, revealing a significant increase in voltage drop ($0.042\text{ V @ }54\text{ mA/cm}^2$) with increased cathode recirculation ratio and reduced fresh air flow. Zhao et al. [43] conducted a detailed orthogonal experiment to explore the humidification capability and voltage clamping effect of cathode recirculation. They found that it increased relative humidity at the cathode inlet, promoting uniform oxygen distribution while adjusting cell voltage to suppress high potential (from 0.828V to 0.780V) at low current density, thereby improving fuel cell stack durability. Liu et al. [44] focused on a dual injector recirculating PEMFC stack, showing that the dual recirculation mode mitigated gas shortage issues during dynamic loading, offering relief from carbon corrosion problems compared to dead anode and dead cathode modes. Zhang et al. [45] investigated humidification and pressure-limiting effects using simulation and experimental methods. High-frequency impedance results indicated a significant reduction in membrane impedance values and oxygen concentration with cathode recirculation. Liu et al. [46] delved into control issues in high-power automotive PEMFC systems, designing an efficient adaptive controller for precise voltage regulation by controlling pump speed. Tu et al. [47] investigated the alleviating effect of cathode recirculation on oxygen starvation under high current density. The results indicated that exhaust gas recirculation significantly mitigated gas starvation and flooding during cycling loads, serving as a crucial alleviation strategy to improve the dynamic characteristics of PEMFC.

1.3. Innovations

Based on the literature review above, it is evident that cathode recirculation can achieve high potential suppression by adjusting the inlet oxygen concentration, thereby alleviating the irreversible degradation caused by carbon corrosion in automotive PEMFC under low-load conditions. While previous research has yielded rich results, there are still many gaps to be filled. For example, the impact of cathode recirculation on PEMFC operation lacks in-depth and effective characterization, especially in terms of characterizing the influence mechanism from an internal dynamics perspective. Additionally, to ensure the effectiveness of cathode recirculation, addressing the control problem of maintaining smooth voltage output and stable tracking against load disturbances is crucial, particularly with efficient and accurate control algorithms. Therefore, based on the points mentioned above, the main innovations of this paper are as follows:

- (1) Investigating the impact of cathode recirculation on three internal polarization processes of PEMFC, revealing the internal influence mechanism of cathode recirculation.
- (2) Establishing a fuel cell system control model that includes a cathode recirculation loop and analyzing its nonlinear characteristics.
- (3) Proposing a multi-model scheduling scheme based on T-S fuzzy theory and exploring the optimal control strategy for high potential suppression under vehicle operating conditions.

The purpose of this paper is to obtain an optimal control scheme to solve the high voltage suppression problem and to provide a new perspective for the durability control of vehicle PEMFC systems under different operating conditions.

2. Experiment setup and PEMFC system model

2.1. Experiment setup of single cell

A 25 cm² commercial membrane assembly (MEA) is used in this study to investigate the effect of cathode circulation on PEM fuel cell performance. The Pt loading of the anode is 0.1 mg/cm² and the Pt loading of the cathode is 0.3 mg/cm². The flow channels at both sides are serpentine channels in parallel wherein the polytetrafluoroethylene (PTFE) gaskets are used for sealing. All single cell experiments are conducted on the Greenlight test station, its actual test bench are shown in Fig. 1 (a). The pure hydrogen is supplied through hydrogen storage tanks, and the compressed air is provided by a compressor. The mass flow rate and inlet gas pressure for the anode and cathode are determined by mass flow controllers and backpressure valves, respectively. The inlet gas can directly be supplied to the single cell or be humidified by the bubble humidifier. The temperature of cell is adjusted by the water bath, and a thermocouple placed at the cathode provides the real-time temperature measurement, which could be regarded as whole-cell temperature during the operation.

In this single cell experiment, cathode circulation ratio (φ) is defined as the ratio of the circulating gas mass flow to the fresh air mass flow. The φ is controlled by increasing the circulating gas and decreasing the fresh air stoichiometric ratio (λ_0) while keeping the total gas volume constant. The experimental pressure of cathode and anode is fixed at 60 °C. The inlet gas pressure for anode and cathode are both controlled

at 100 kPa using the back pressure valve. λ_0 is controlled at 3.0 and hydrogen mass flow is kept at 0.07SLPM. The electrochemical workstation, Gamry Interface 5000E, is used to measure EIS. The frequency range in this study is from 10 kHz to 0.1 kHz, and the amplitude is picked to be 8% of DC load to achieve a linear measure.

2.2. PEMFC system dynamic model

In this section, a physical model of a typical vehicle PEMFC system equipped with a cathode recirculation loop is developed for use in the development of idle controllers to mitigate the effects of high potentials. The test bench of the automotive PEMFC system is shown in Fig. 1(b), through which the accuracy of the established model is verified by collecting operational data.

2.2.1. Electrochemical model

The accurate modeling of the voltage of the fuel cell is crucial as it is directly related to the evaluation of the control effect. The electrochemical model can be modeled as:

$$V_{fc} = N \cdot V_{cell} = N \cdot (V_{nernst} - V_{act} - V_{ohm} - V_{con}) \quad (1)$$

where, V_{fc} is the stack voltage, V_{cell} is the individual voltage, N is the number of individual cells, V_{nernst} represents the thermodynamic Nernst voltage, V_{act} denotes the activation loss, V_{ohm} is the ohmic loss, V_{con} is the concentration loss. V_{nernst} can be expressed as:

$$V_{nernst} = E_0 - \frac{\Delta s}{2F} (T_{st} - T_{ref}) + \frac{RT_{st}}{2F} \left[\ln(p_{H_2}) + \frac{1}{2} \ln(p_{O_2}) \right] \quad (2)$$

where E_0 represents the standard electric potential, p_{H_2} is the hydrogen partial pressure, p_{O_2} is the oxygen partial pressure, T_{st} represent the cell

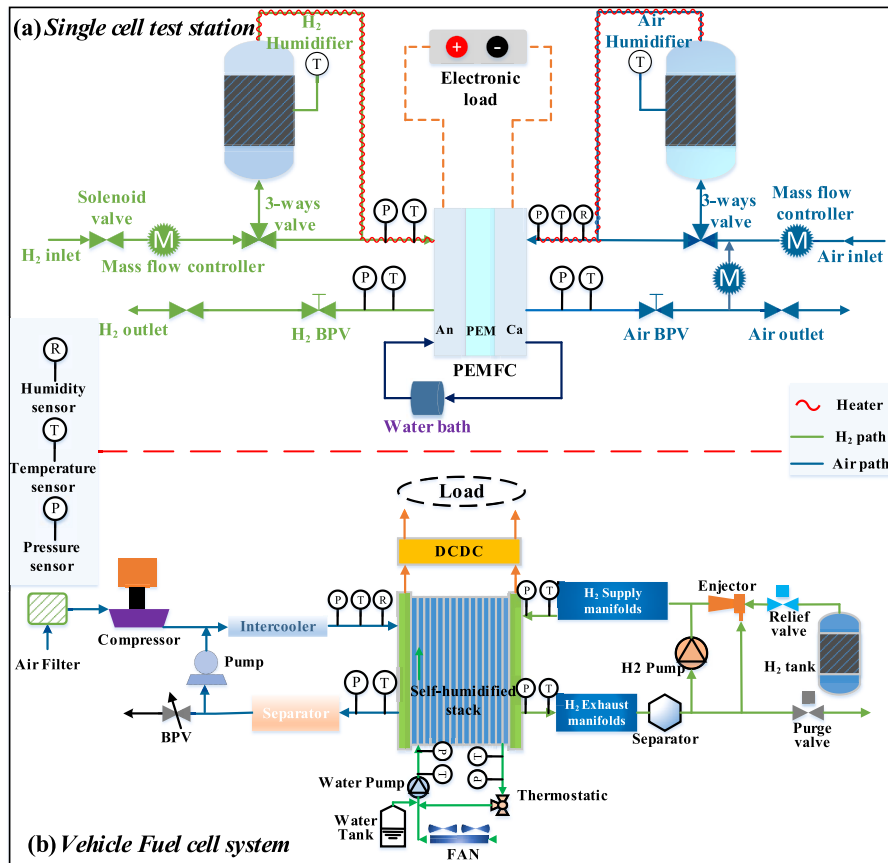


Fig. 1. Cathode circulation experimental bench and its vehicle system. (a) G20 test station, (b) Integrated system bench for vehicle applications.

temperature. In addition, the three voltage losses can be calculated as:

$$\begin{cases} V_{act} = k_1 + k_2 T_{st} + k_3 T_{st} \ln(C_{O_2}) + k_4 T_{st} \ln(i * A) \\ V_{ohm} = i \times R_{ohm} \\ V_{con} = c \ln\left(i_{max}/(i_{max} - i)\right) \end{cases} \quad (3)$$

where $k_1 \sim k_4$ are the identification parameters, c is the empirical coefficient, i is the current density, A is the activation area, i_{max} is the limiting current density. C_{O_2} is the dissolved oxygen concentration at the cathode reaction interface, which can be expressed as:

$$C_{O_2} = 1.969 \times 10^{-7} p_{O_2} \exp(-T_{st}/498) \quad (4)$$

R_{ohm} is the total resistance can be expressed as:

$$R_{ohm} = \frac{l_m}{(k_5 \lambda_m + k_6) \exp\left(k_7 \left(\frac{1}{303} - \frac{1}{T_{st}}\right)\right)} \quad (5)$$

where, l_m is the thickness of the membrane, $k_5 \sim k_7$ are the identification parameters, λ_m is the average water content.

Equations (1)–(5) constitute the complete fuel cell electrochemical semi-empirical and semi-mechanical model, in which the Nernst voltage and activation losses are directly influenced by oxygen pressure; thus the proposed cathode recirculation will take the reacted exhaust gas refeed to the inlet to achieve the purpose of voltage regulation. The parameters in the voltage model are collected in Table 1.

2.2.2. Cathode/anode mass balance

According to mass conservation and the ideal gas law, the internal gas pressure at the cathode/anode of the fuel cell is determined by the inlet and outlet flow rates and the electrochemical reaction rate. The partial pressure of the gas at the cathode is modeled as follows:

$$\frac{dp_{O_2}}{dt} = \frac{R_{O_2} T_{st}}{V_{ca}} (W_{O_2,in} - W_{O_2,out} - W_{O_2,react}) \quad (6)$$

$$\frac{dp_{N_2}}{dt} = \frac{R_{N_2} T_{st}}{V_{ca}} (W_{N_2,in} - W_{N_2,out}) \quad (7)$$

The oxygen consumption can be calculated from the stack current I_{st} .

$$W_{O_2,react} = \frac{NI_{st}}{4F} \times M_{O_2} \quad (8)$$

The partial pressure of water vapor inside the cathode is:

$$\begin{cases} \frac{dp_{v,ca}}{dt} = \frac{R_{H_2O} T_{st}}{V_{ca}} (W_{v,in,ca} - W_{v,out,ca} + W_{v,gen} + W_{v,mem}) & \text{if } p_{v,ca} < p_{sat}(T_{st}) \\ p_{v,ca} = p_{sat}(T_{st}) & \text{if } p_{v,ca} \geq p_{sat}(T_{st}) \end{cases} \quad (9)$$

Table 1

Calibration values of PEMFC electrochemical model.

Symbol	Description	Value	Unit
N	Cell number	360	/
A	Fuel cell active area	300	cm ²
F	Faraday constant	96,485	C/mol
c	empirical constant	0.02	/
i_{max}	Limiting current density	2.2	A/cm ²
k_1	identified coefficient	0.1406	/
k_2	identified coefficient	-8.4×10^{-3}	/
k_3	identified coefficient	-5.01×10^{-4}	/
k_4	identified coefficient	4.31×10^{-5}	/
k_5	identified coefficient	-1.34×10^{-2}	/
k_6	identified coefficient	0.2472	/
k_7	identified coefficient	0.5746	/

The rate of water production is:

$$W_{v,gen} = \frac{NI}{2F} \times M_{H_2O} \quad (10)$$

In equations 6–10, N is the number of cells, V_{ca} is the cathode volume, p_{N_2} is the partial pressure nitrogen in the cathode, R_{O_2} and R_{N_2} are the gas constant for oxygen and nitrogen, $W_{O_2,in}$ and $W_{O_2,out}$ are the mass flow rate of oxygen into and out of the cathode, $W_{N_2,in}$ and $W_{N_2,out}$ are the mass flow rate of nitrogen into and out of the cathode, $p_{v,ca}$ is the partial pressure of water vapor at the cathode, M_{O_2} is the molar mass of oxygen, R_{H_2O} and M_{H_2O} are the gas constant and molar mass of water vapor, $W_{v,in,ca}$ and $W_{v,out,ca}$ are the inflow flow rate and outflow flow rate of water at the cathode, $W_{v,gen}$ and $W_{v,mem}$ are water production rate and water flow rate through the membrane, $p_{sat}(T_{st})$ is the saturation vapor pressure.

The hydrogen partial pressure and water vapor partial pressure in the anode channel can be modeled as follows:

$$\frac{dp_{H_2}}{dt} = \frac{R_{H_2} T_{st}}{V_{an}} (W_{H_2,in} - W_{H_2,out} - W_{H_2,react}) \quad (11)$$

$$\begin{cases} \frac{dp_{v,an}}{dt} = \frac{R_{H_2O} T_{st}}{V_{an}} (W_{v,in,an} - W_{v,out,an} - W_{v,mem}) & \text{if } p_{v,an} < p_{sat}(T_{st}) \\ p_{v,an} = p_{sat}(T_{st}) & \text{if } p_{v,an} \geq p_{sat}(T_{st}) \end{cases} \quad (12)$$

The flow rate of hydrogen gas consumed by the electrochemical reaction is:

$$W_{H_2,react} = \frac{NI_{st}}{2F} \times M_{H_2} \quad (13)$$

In equations 11–13, V_{an} , R_{H_2} and M_{H_2} are the anode volume, the gas constant and the molar mass of the hydrogen, respectively. p_{H_2} and $p_{v,an}$ are hydrogen partial pressure and water vapor partial pressure at the anode, respectively. $W_{H_2,in}$, $W_{H_2,out}$ and $W_{H_2,react}$ are the inlet flow rate, the outlet flow rate and the reaction consumption flow rate of the anode hydrogen, respectively. $W_{v,in,an}$ and $W_{v,out,an}$ are the inlet flow and outlet flow of the water vapor at anode, respectively.

2.2.3. Air compressor model

The model of the air compressor consists of speed dynamic and static map. In a large number of previous research works, the dynamic of the air compressor is modeled as a lumped rotational parameter model, where the manipulated variable motor voltage. However, current commercial PEMFC systems support air compressors with speed as the only manipulated variable. Therefore, in this paper, the rotational speed, which is closer to the actual application, is used as the only control input for the air compressor, which facilitates the modeling and verification of the flow MAP. The dynamic of the speed is concisely modeled as a one-order inertia link:

$$n_{actual,cp}(s) = \frac{1}{\tau_{cp}s + 1} n_{cp}(s) \quad (14)$$

where n_{cp} represents the input speed of the air compressor. $n_{actual,cp}$ is the actual speed considering the time lag, τ_{cp} is the time constant.

The static map of the compressor is expressed by collecting experimental data with a centrifugal compressor and training a radial basis function (RBF) forward neural network model, which improves the fidelity of the model, instead of the previous empirical formula method. The inputs to the RBF model are the pressure ratio (p_{sm}/p_{atm}) and the actual compressor speed. p_{sm} is the supply manifold pressure determined by the manifold dynamics, p_{atm} is the atmospheric pressure. The output is the mass flow rate (W_{cp}) of the compressor. The validation of the RBF model based on the Gaussian kernel function is given in Section 4.2.

2.2.4. Circulation pump model

The control model of the cathode circulation pump is referenced to the modeling process of the air compressor. The only manipulated variable of the pump is the speed, the dynamics of which is simulated by the one-order inertia link:

$$n_{actual,p}(s) = \frac{1}{\tau_p s + 1} n_{pump}(s) \quad (15)$$

where n_{pump} represents the input speed of the pump, $n_{actual,p}$ is the actual speed, τ_p is time constant.

The operating of the circulating pump selected in the actual system determine the flow rate as a function of the speed and differential pressure. A flow rate model based on a back propagation (BP) neural network is developed by collecting experimental data from the pump. The inputs to the model are the pump speed and the differential pressure ($p_{sm} - p_{rm}$), p_{rm} is the exhaust pressure determined by the manifold dynamics, and the output is the pump mass flow rate (W_{pump}). The validation of the flow model for the BP artificial neural network is given in Section 4.2.

In the section of pump and compressor modeling, a data-driven approach is used instead of the traditional first-principles-based modeling approach, which improves model fidelity, cuts the model's parametric complexity, and increases the speed of real-time calculations.

2.2.5. Manifold dynamics

According to the conservation of mass, the pressure dynamics of the supply and exhaust manifolds can be calculated as follows:

$$\frac{dp_{sm}}{dt} = \frac{T_{mix} R_{air}}{V_{sm}} (W_{sm,in} - W_{sm,out}) \quad (16)$$

$$\frac{dp_{rm}}{dt} = \frac{T_{st} R_{air}}{V_{rm}} (W_{rm,in} - W_{rm,out}) \quad (17)$$

where V_{sm} and V_{rm} are the volumes of the inlet and exhaust manifolds, respectively. T_{mix} is the temperature of the mixed gases from the compressor and the circulation pump. $W_{sm,in}$, $W_{sm,out}$, $W_{rm,in}$, $W_{rm,out}$ are the supply manifold inlet and outlet flows and the exhaust manifold inlet and outlet flows, respectively. Due to the small pressure difference, $W_{sm,out}$ and $W_{rm,in}$ are:

$$W_{sm,out} = k_{sm,out} (p_{sm} - p_{ca}) \quad (18)$$

$$W_{rm,in} = k_{rm,in} (p_{ca} - p_{rm}) \quad (19)$$

where, $k_{sm,out}$ and $k_{rm,in}$ are the flow coefficients of the cathode inlet and outlet, respectively.

Calculating the mass flow at the exhaust manifold outlet by means of the nonlinear nozzle flow equation:

$$W_{rm,out} = \begin{cases} \frac{C_D \pi D^2 (1 - \cos(\theta)) p_{rm}}{4 \sqrt{RT_{st}}} \left(\frac{p_{atm}}{p_{rm}} \right)^{\frac{1}{\gamma}} \sqrt{\frac{2\gamma}{\gamma-1} \left[1 - \left(\frac{p_{atm}}{p_{rm}} \right)^{\frac{\gamma-1}{\gamma}} \right]} & \frac{p_{atm}}{p_{rm}} > \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} \\ \frac{C_D \pi D^2 (1 - \cos(\theta)) p_{rm}}{4 \sqrt{RT_{st}}} \sqrt{\gamma \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{(\gamma-1)}}} & \frac{p_{atm}}{p_{rm}} \leq \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} \end{cases} \quad (20)$$

where, C_D is the charge coefficient, D and θ are the diameter and opening angle of the nozzle, respectively. R is the universal gas constant.

Due to the addition of the dynamic process of cathode recirculation in the air subsystem, $W_{sm,in}$ is approximated by the sum of:

Table 2

Parameters and constants for model simulation.

Symbol	Description	Values	Unit
V_{sm}	Supply manifold volume	0.02	m ³
V_{rm}	Return manifold volume	0.005	m ³
V_{ca}	Cathode volume	0.01	m ³
V_{an}	Anode volume	0.005	m ³
$k_{sm,out}$	Cathode inlet flow constant	2.2×10^{-4}	kg/(s*Pa)
$k_{rm,in}$	Cathode outlet flow constant	3.6×10^{-4}	kg/(s*Pa)
γ	Ratio of specific heat of air	1.4	/
C_D	Discharge coefficient of nozzle	0.012	/
R	Universal gas constant	8.3415	J/(mol*K)
R_{O_2}	Oxygen gas constant	259.8	J/(kg*K)
R_{N_2}	Nitrogen gas constant	296.8	J/(kg*K)
R_{H_2O}	Vapor gas constant	461.5	J/(kg*K)
M_{O_2}	Oxygen molar mass	32	g/mol
M_{N_2}	Nitrogen molar mass	28	g/mol
M_{H_2O}	Vapor molar mass	18	g/mol
p_{atm}	Atmosphere pressure	101.325	kPa
M_{H_2}	Hydrogen molar mass	2	g/mol
R_{H_2}	Hydrogen gas constant	4.124	J/(kg*K)

$$W_{sm,in} = W_{cp} + W_{pump} \quad (21)$$

Equations 1–21 form a nonlinear model of the PEMFC and its air supply subsystem, including the air supply, electrochemical reaction, and exhaust cycle. The model simulation is executed in the MATLAB/Simulink environment, which can simulate a series of dynamic processes. Other auxiliary processes such as hydrogen supply, air-exhaust gas mixing, etc. are not described individually and can be referred to our previous work [46]. The parameters and constants used for the model simulation other than the electrochemical reactions are collected in Table 2.

3. Methodology

3.1. Problem definition

Section 2 develops a novel dynamic fuel cell system model with the cathode recirculation process. The purpose of cathode recirculation is applied to regulate the voltage value at low load conditions. In this work, the controlled variable is the output voltage. To ensure the system's reliability, stability and durability, the output voltage must follow the set value precisely, even when subject to strong disturbances. Therefore, manipulated variables and controllers are critical. Generally, continuous or cyclic operation above 0.8v is unacceptable in low/idle control of vehicle fuel cells, so 0.8v is set as the control target.

In the control system of this paper, the speed of the recirculation pump is set as the control input (u), the current can be considered as disturbance (w) and the voltage is the measurement output (y). Due to the concern of high voltage control under low load conditions, offline

calibration of the air compressor is carried out, and the OER is maintained at the set value. The proportional controller for the air compressor speed in response to load variation is as follows:

$$n_{cp} = 432.4 \times I_{st} + 33130 \quad (22)$$

3.2. Linearization

Compared to nonlinear model-based control, control methods based on nominal linearized models have the advantages of computational simplicity and good real-time performance easily implemented in commercial controllers; however, the key to such methods is to obtain dynamic characteristics that are highly compatible with nonlinear systems. This section linearizes the nonlinear model established in Section 2 and discusses the dynamic characteristics of the linearized model at different nominal points.

Due to the actual system's operational behavior, where the voltage exceeds 0.8V when the current is below 80A, three nominal operating points were selected for linearization at 40A, 50A, and 70A. The linear model is

$$\begin{cases} \delta\dot{\mathbf{x}} = A\delta\mathbf{x} + B_u\delta\mathbf{u} + B_w\delta\mathbf{w} \\ \delta\mathbf{y} = C\delta\mathbf{x} + D_u\delta\mathbf{u} + D_w\delta\mathbf{w} \end{cases} \quad (23)$$

Where, $\delta(\cdot)$ represents the variation from the nominal value. The state, $\mathbf{x}$, measurement output, $\mathbf{y}$, input, $\mathbf{u}$, and disturbance, $\mathbf{w}$, are

$$\begin{cases} \mathbf{x} = [p_{rm} \ n_{pump} \ n_{cp} \ p_{sm} \ p_{H_2} \ m_{w,an} \ p_{N_2} \ p_{O_2}]^T \\ \mathbf{y} = [V_{cell} \ p_{sm} \ p_{rm}]^T \\ \mathbf{u} = n_{pump} \\ \mathbf{w} = I_{st} \end{cases} \quad (24)$$

The linearized model comprises 8 states, 2 inputs (control input and disturbance input), and 3 outputs. The operating conditions of the system at the three nominal operating points are given in Table 3.

Fig. 2 illustrates the dynamic characteristics of the linearized model at three nominal operating points. The results show that the dynamics of the models are different at different nominal points, which represents the effect of nonlinearity. Meanwhile, when the step signal is near the nominal point, the nominal model can highly match the dynamic characteristics of the nonlinear model, however, as the step signal moves away from the nominal point, the problem of mismatch appears. This is due to the oxygen partial pressure dynamics that determines the output voltage is influenced not only by the circulation pump but also by the air compressor, and the coupling of the two increases the nonlinearity of the system, which leads to the mismatch caused by the approximate linearization. Therefore, the control error caused by the model mismatch needs to be avoided when using the control design based on the linearized model.

3.3. T-S fuzzy theory

To address model mismatch, we employ a multi-model fusion approach. T-S fuzzy theory, grounded in a multi-model scheduling methodology, utilizes fuzzy rules to precisely approximate a linearized model to a nonlinear one, effectively mitigating nonlinearity. In this study, fuzzy rules are devised for the three nominal models outlined in Section 3.2, incorporating current as the fuzzy variable. This is formulated as:

Rule i:

If u_1 is M_{i1} , ..., and μ_g is M_{ig} .

Then $\dot{\mathbf{x}} = A_i\mathbf{x}(t) + B_i\mathbf{u}(t)$, $i = 1, 2, 3, \dots, r$

Table 3

The operating conditions of the three nominal models.

Model	I_{st}/A	n_{cp}/rpm	n_{pump}/rpm	T_{st}/K	$\theta/^\circ$	V_{cell}/V
M ₁	40	45,000	8500	338	30	0.8
M ₂	50	50,000	7600	338	30	0.8
M ₃	70	60,000	4800	338	30	0.8

Where, r is the number of fuzzy rules, u is premise variables, in this study is I_{st} , M is the fuzzy set.

The constraint of fuzzy rules makes the weighted output of multiple linear models, thus improving the control effect over a wide range of operation. The output based on T-S fuzzy theory can be expressed as:

$$y(k) = \sum_{i=1}^n \alpha_i \cdot y_i(k) \quad (25)$$

where, k is the sample moment, i is the model number, α_i is the weight of the i -th nominal model, $y_i(k)$ is the output of the i -th model, $y(k)$ is the total output.

3.4. LQG design

According to Equation (23), the state space expression from the input, $\mathbf{u}$, to the first measurement variable, δy_1 (V_{cell}), is given by:

$$\begin{cases} \delta\dot{\mathbf{x}} = A\delta\mathbf{x} + B_u\delta\mathbf{u} \\ \delta y_1 = C_{y1}\delta\mathbf{x} + D_w\delta\mathbf{w} \end{cases} \quad (26)$$

A linear quadratic regulator (LQR) optimal control is used to design a state feedback controller based on output regulation. Since the LQR problem is concerned with the convergence of states and inputs, the cost function is in the form

$$J = \int_0^\infty \delta\mathbf{x}^T C_{y1}^T Q_y C_{y1} \delta\mathbf{x} + \delta\mathbf{u}^T R \delta\mathbf{u} \quad (28)$$

where Q_y is the tuning matrix of the output, R is the tuning matrix of the input, they are responsible for the state tracking or control energy priorities and need to be chosen reasonably.

LQR control aims to compute a set of optimal control actions that minimize the cost function is given by

$$\delta\mathbf{u} = -K\delta\hat{\mathbf{x}}, K = R^{-1}(B_u^T S + N^T) \quad (29)$$

When measurements from different sensors are available but subject to noise, Kalman filter combines sensory data from various sources (known as sensor fusion) to find the best estimate of the parameter of interest. To simulate real experimental conditions, process and measurement white noise are added to system and the state space expression of the system is rewritten as:

$$\begin{cases} \delta\dot{\mathbf{x}} = A\delta\mathbf{x} + B_u\delta\mathbf{u} + B_w\delta\mathbf{w} + G\delta\mathbf{w}_1 \\ \delta\mathbf{y} = C_y\delta\mathbf{x} + D_u\delta\mathbf{u} + D_w\delta\mathbf{w} + \mathbf{v} \end{cases} \quad (30)$$

where, G is a noise transformation matrix. Both $\mathbf{w}_1$ and $\mathbf{v}$ are modeled as Gaussian white noise.

The Kalman filter calculates the feedback matrix K_k and completes the optimal state estimation, which is a two-part process, prediction and correction [23]. Based on the theory of the observer, the state and output equations of the Kalman estimator are:

$$\delta\dot{\hat{\mathbf{x}}} = A\delta\hat{\mathbf{x}} + B_u\delta\mathbf{u} + B_w\delta\mathbf{w} + K_k(\delta\mathbf{y} - C_y\delta\hat{\mathbf{x}} - D_u\delta\mathbf{u} - D_w\delta\mathbf{w}) \quad (31)$$

$$\begin{bmatrix} \delta\hat{\mathbf{y}} \\ \delta\hat{\mathbf{x}} \end{bmatrix} = \begin{bmatrix} C_y \\ I \end{bmatrix} \delta\hat{\mathbf{x}} + \begin{bmatrix} D_u \\ 0 \end{bmatrix} \delta\mathbf{u} + \begin{bmatrix} D_w \\ 0 \end{bmatrix} \delta\mathbf{w} \quad (32)$$

The Kalman filter uses the known inputs $\mathbf{u}$, $\mathbf{w}$ and the measurements $\mathbf{y}$ to generate the estimated output $\hat{\mathbf{y}}$ and state $\hat{\mathbf{x}}$. Based on equation (32), The Kalman filter is a state-space model having 5 inputs and 11 outputs. The input variables are $[I_{st} \ n_{pump}]^T$ and $[V_{cell} \ p_{sm} \ p_{rm}]^T$ with noisy. The first three outputs are the estimated true system output $\hat{\mathbf{y}}$, the remaining eight outputs are the state estimates $\hat{\mathbf{x}}$.

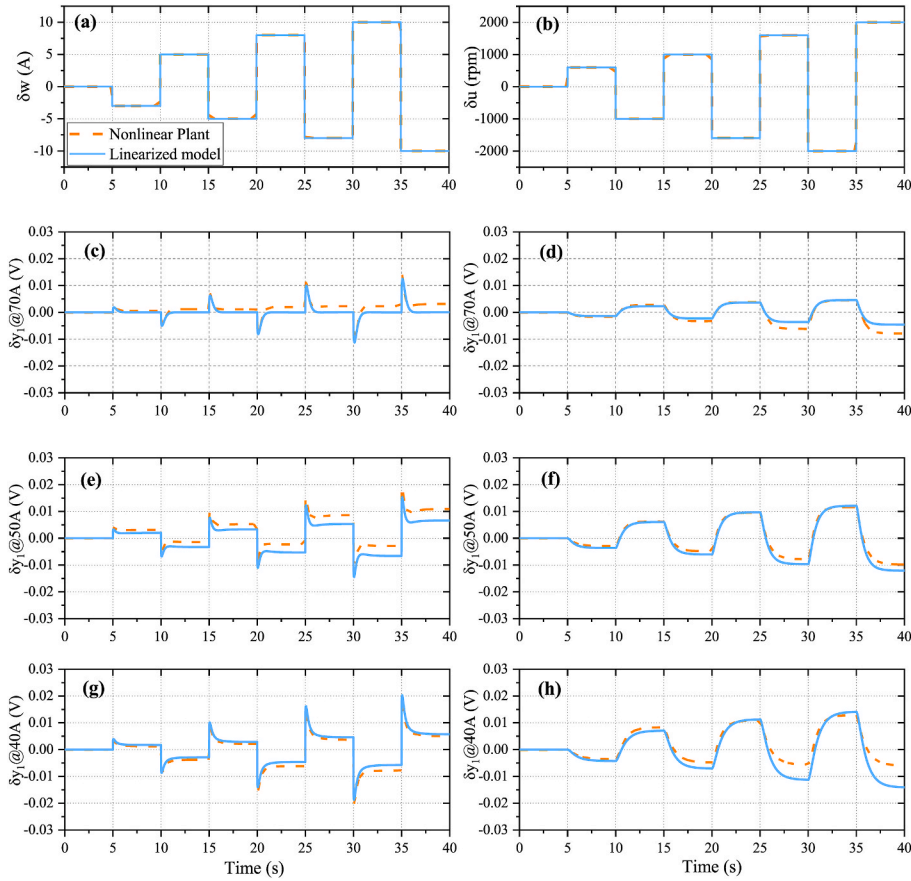


Fig. 2. Comparison of step response of nominal model and nonlinear plant.(a) Variation of δw , (b) Variation of δu ,(c),(e) and (g) represent the variation of output δy_1 at different current points with δw , (d),(h) and (f) represent the variation of output δy_1 at different current points with δu .

3.5. MPC design

The MPC mechanism is described as follows: at each sampling moment, a finite-time open-loop optimization problem is solved online based on the current measurement information and the obtained first element of the resulting control sequence is applied to the controlled object. The MPC algorithm consists of three steps: prediction, optimization, and feedback correction. The linear state space shown in Equation (26) is used to design MPC controller. The discrete predicted system incremental model is shown in Equation (33). The predicted p-step state in MPC is shown in Equation (34).

$$\begin{cases} \delta x(k+1) = \varphi \delta x(k) + G \delta u(k) \\ \delta y_1(k) = C_{y1} \delta x(k) + D_w \delta w(k) \end{cases} \quad (33)$$

$$\delta x^p(k+p|k) = \varphi^p \delta x(k) + G^p \delta U(k) \quad (34)$$

Where $\varphi = I + kA$, $G = TB_u$ and k is the sampling moment.

$$\varphi^p = \begin{bmatrix} I \\ \varphi \\ \varphi^2 \\ \vdots \\ \varphi^p \end{bmatrix}, G^p = \begin{bmatrix} 0 & 0 \cdots 0 \\ G & \cdots \\ \varphi G & G \\ \vdots & \vdots \\ \varphi^{p-1} G & \varphi^{p-2} G \cdots G \end{bmatrix}, \delta U(k) = \begin{bmatrix} \delta u(k|k) \\ \delta u(k+1|k) \\ \vdots \\ \delta u(k+p|k) \end{bmatrix}$$

In this study, we expect output voltage can follow the reference signal quickly and accurately with only minimal control cost. Therefore, the objective function and constraints can be expressed as follows:

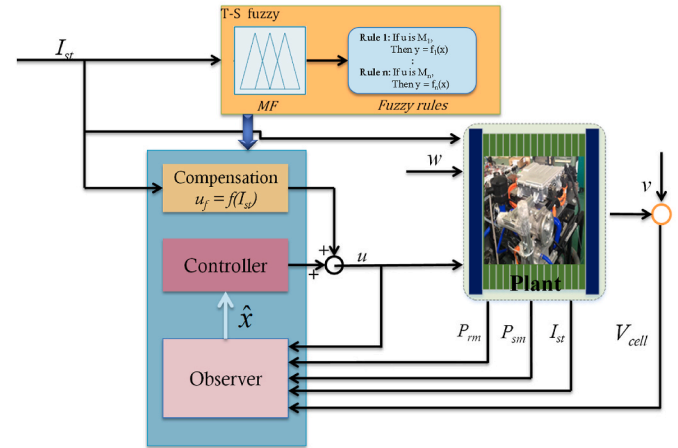


Fig. 3. The proposed optimal control scheme in this paper.

$$J = \gamma \sum_{j=N_1}^{N_2} (y_r(k+j) - y_m(k+j))^2 + \rho \sum_{j=1}^{N_u} (\dot{u}'(k+j-1) - \dot{u}'(k+j-2))^2$$

$$s.t. \begin{cases} 10^3 \leq u(k+j) \leq 10^4, j = 1 \cdots N_u \\ |\delta u(k+j-1)| \leq 1000, j = 1 \cdots N_u \end{cases} \quad (35)$$

where, N_1 and N_2 represent the start and end time of the prediction horizon. N_u define the time domain of the control increments. $\dot{u}'$ is the

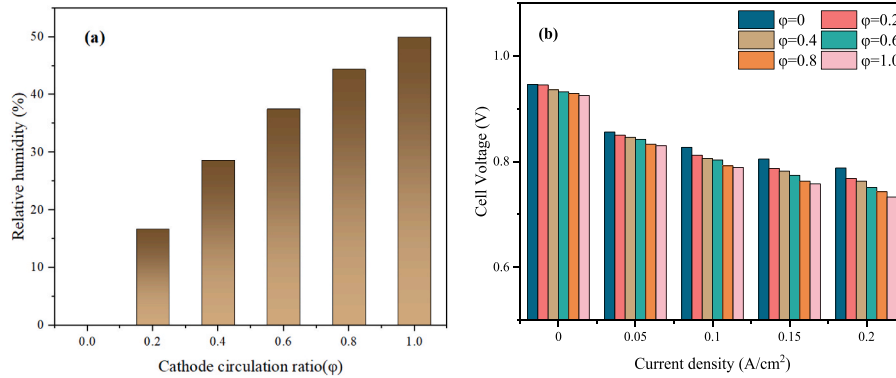


Fig. 4. Effect of cathode circulation ratio on PEMFC external performance. (a) Relative humidity at cathode inlet. (b) Variation of output voltage.

tentative control signal, y_r is the desired voltage response, and y_m is the model response of voltage. γ and ρ are the weight metrics.

In this chapter, MATLAB/Simulink serves as the computational platform for model linearization, control algorithm design, and solution. The involved toolboxes include the Control System Design Toolbox, Statistics and Machine Learning Toolbox, and Predictive Control Toolbox, among others. Simulation and comparative analysis of different algorithms are conducted in the Simulink environment. The control system structure based on the TS fuzzy model fusion optimal control algorithm is visualized in Fig. 3.

4. Results and discussion

4.1. Characteristics analysis of cathode cycling on PEMFC

Fig. 4(a) shows the variation in relative humidity at cathode inlet with increasing ϕ . It can be found that the humidity tends to increase gradually as ϕ increases. When no circulation is applied, the air is set to dry air, but when the circulation ratio reaches 1.0, the humidity is around 50%. It is obvious that the cathode cycle brings a substantial amount of water vapor and thus humidifies the intake air. Fig. 4(b) shows the voltage suppression effect under different current and ϕ . In this case, the fresh air stoichiometry without circulation is set to 3.0. As the ϕ increases, the voltage tends to decrease at different current densities. This is due to the increase in circulating gas, which reduces the partial pressure of oxygen in the cathode intake, resulting in voltage suppression. Moreover, voltage is more sensitive to changes in the ϕ at high currents than at low currents. Specially, as the current density increases, the voltage drops are 0.021V, 0.026V, 0.038V, 0.047V and 0.055V via an increase in the cyclic ratio from 0 to 1.0, respectively. The rate of decrease is 2.2%, 3%, 4.6%, 5.8% and 7%, respectively.

At present, 0.8V is commonly adopted as the control threshold,

Table 4

The key parameter settings in DRTtools.

Items	Values
Method of discretization	Gaussian
Regularization derivative	1st order
Regularization parameter	1e-5
RBF shape control	FWHM coefficient
FWHM coefficient	0.5

hence, the adjustable current region occurs at 0.05–0.15A/cm². The optimal ϕ are 1.0, 0.8, and 0.2, which obviously should be appropriately reduced with increasing current.

Fig. 5(a) shows the effect of the ϕ on the internal polarization process when current density is 0.1A/cm². In this paper, EIS data were fitted using DRTtools [48]. Detailed parameter settings are shown in Table 4. It can be found from Fig. 5(a) that the impedance spectral radius gradually increases and shifts to the left with increasing ϕ . The semicircle of the middle frequency is much larger than the semicircle of the high frequency, which suggests that the internal polarization at this stage is dominated by the oxygen reduction reaction (ORR) at cathode. With the increase of ϕ , the cathode oxygen partial pressure as well as the oxygen concentration decreases gradually, which reduces the ORR reaction and oxygen diffusion rate, leading to an increase in impedance. The change from the internal polarization process also directly explains the fact that the external voltage is reduced in Fig. 4(b). Conversely, ohmic internal resistance demonstrates the opposite trend. The ohmic loss in the Nyquist plot is described as high-frequency intercept with the real axis. The leftward shift in the impedance spectrum indicates a gradual decrease in ohmic resistance with increasing ϕ . This is because the increase in the ϕ raises the water content within the membrane, which reduces the resistance to proton conduction as well as the contact

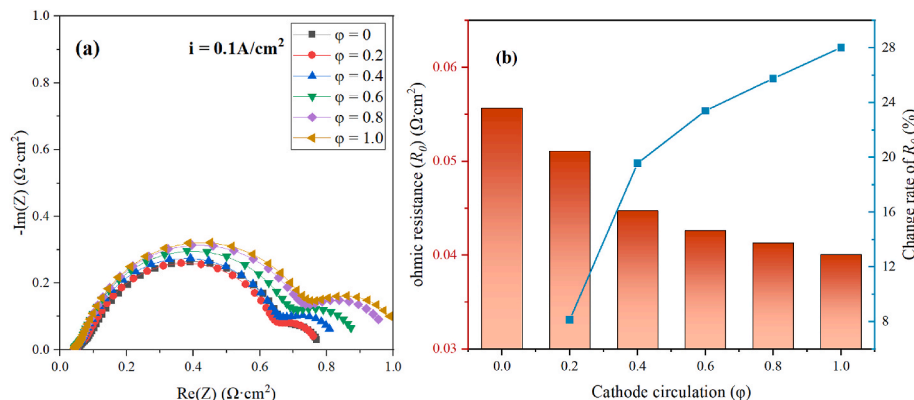


Fig. 5. Effect of cathode circulation ratio on PEMFC internal polarization process. (a) Electrochemical impedance spectroscopy, (b) Ohmic resistance.

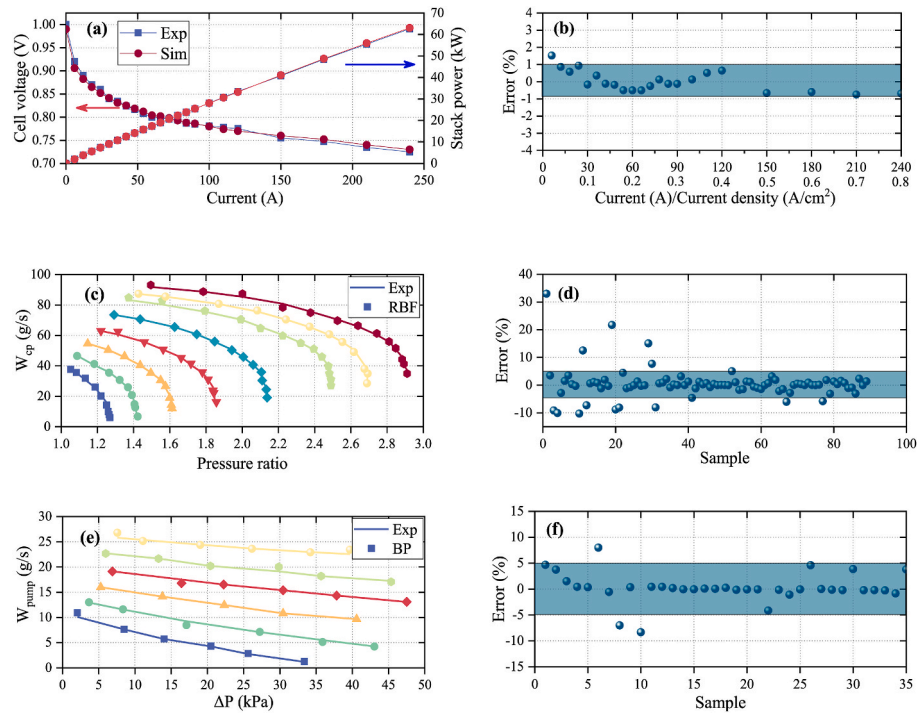


Fig. 6. Validation of the PEMFC system dynamic model. (a-b) Verification and error of the electrochemical model, (c-d) Verification and error of the RBF model for the air compressor flow rate, (e-f) Verification and error of the BP model for the recirculation pump flow rate.

resistance between the membrane and the electrodes. Fig. 5(b) further quantifies the change in ohmic resistance with different ϕ . As ϕ increases, the ohmic internal resistance decreases from $0.055 \Omega \text{ cm}^2$ to $0.04 \Omega \text{ cm}^2$, a reduction of 28.1%. However, it is also observed that the rate of decrease slows down with increasing ϕ , which corresponds to the

external humidification results shown in Fig. 4 (a). In Figs. 4 and 5, the changes in the performance of the PEMFC due to cathode cycling are analyzed from an internal plus external perspective, which provides a theoretical basis for the design of controllers.

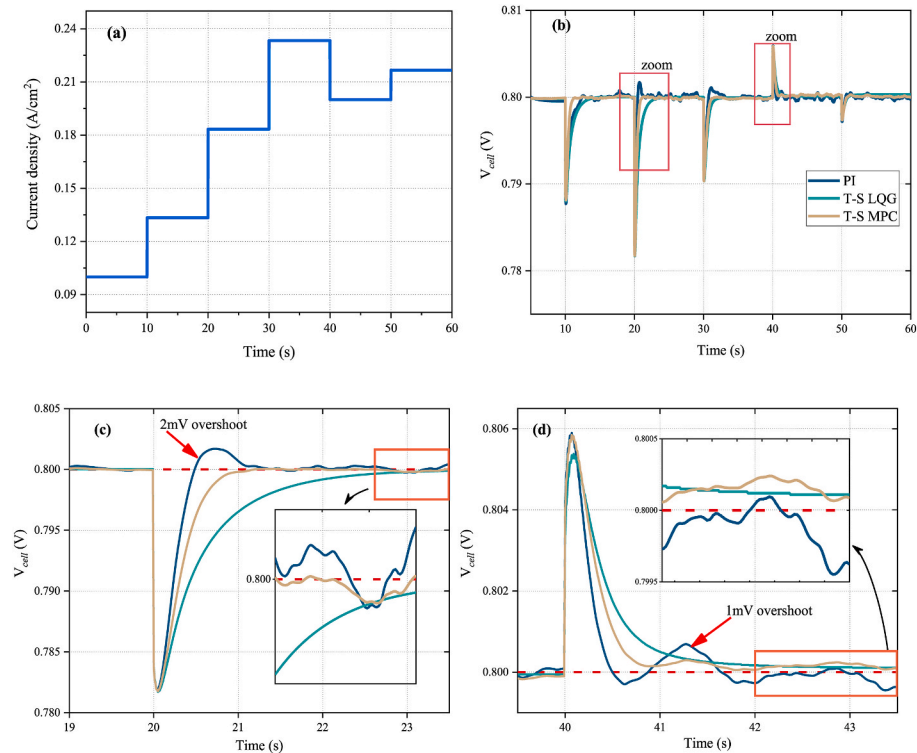


Fig. 7. Comparison of the voltage control effect of different control method. (a) is current density profile, (b) is voltage control effect, (c) and (d) are the enlarged views of the yellow boxes in Fig. 7(b).

4.2. Validation of the PEMFC system model

To bring the control simulation closer to the real experiment, the accuracy of the nonlinear simulation model developed in Section 2 needs to be verified. The validation data comes from the system shown in Fig. 1(b). The polarization curve data are collected through current load experiments on the real fuel cell system, and the voltage model parameters are identified, as shown in Table 1. Fig. 6(a) verifies the steady-state voltage and power output of the model. The current is loaded from 0A to 240A. Since this study is more concerned with the voltage output at low load, a more intensive loading is performed at this condition. The results show that the simulation results agree with the experimental data, with the vast majority of errors within 1% and the largest voltage errors controlled within 2%, demonstrating the high fidelity of the established voltage model.

Furthermore, the flow models of the air compressor and circulation pump, which are essential auxiliary components in the air supply subsystem, are validated and the results are shown in Fig. 6(b) and (c). The results show that the air compressor flow MAP established by BRN neural network is highly consistent with the experimental data, and the relative error of most of the sample points is within $\pm 5\%$. It is worth noting that the extreme error point occurs at the air compressor surge boundary when the flow rate is very low, and small fluctuations can cause great relative error values, so it should not be considered. For the circulation pump, the results show that the relative error is controlled to within $\pm 5\%$. Overall, the model of the voltage and auxiliary components clearly captures the actual system characteristics with errors within acceptable limits for controller development.

4.3. Voltage control effect comparison

The control performance of various controllers is assessed using a current profile depicted in Fig. 7(a), and the results are presented in Fig. 7(b)–(d). The control objective in this study is to achieve stable and fast-tracking voltage at the set point of 0.8V under different current disturbances. Fig. 7(b) illustrates the voltage control effectiveness of different control methods. Without running a cathode cycle, the voltage exceeds 0.8V, which is unacceptable for the long-term operation of the PEMFC. During the tracking phase, the static errors of all three controllers are sufficiently small. However, during load changes, PI controllers inevitably experience overshooting, and the T-S LQG controller based on internal state feedback eliminates overshooting but exhibits a significant lag in dynamic response. In contrast, the T-S MPC method demonstrates the best balance between transient response and steady-state error, attributed to MPC's advanced prediction and real-time optimization mechanism. Fig. 7(c) and (d) provide enlarged views of the control effect during the load up and load down stages, respectively, for a more visual comparison.

Additionally, the performance of the different control methods during dynamic adjustment is quantified in Table 5. In the load-up instant (@20s), the overshoot of PI controller is 2 mV, while the value of T-S LQG and T-S MPC is 0. The turning time of T-S MPC is about 1s, which is 33.3% and 66.7% faster than PI and T-S LQG. Similarly, in the load-down moment (@40s), T-S MPC is 50% and 60% faster than the other two control methods. These results show that the T-S MPC coupled control method has excellent performance against the step current signal.

Table 5
Quantitative analysis results of different control methods.

Methods	Load-up stage		Load-down stage	
	Overshoots	Turning time	Overshoots	Turning time
PI	2 mV	1.5s	1 mV	2s
T-S LQG	0	3s	0	2.5s
T-S MPC	0	1s	0	1s

4.4. Robustness of controllers

In the context of potential unexpected events during fuel cell system operation that could lead to system instability, it is crucial for the designed controller to demonstrate robustness against random disturbances. Within the air supply system, the air compressor significantly influences the fuel cell's voltage output by directly modifying the oxygen concentration entering the stack through its flow. When the air compressor experiences sudden changes, the high-potential controller must swiftly respond to stabilize the voltage at the desired value. In this section, a disturbance of 3% is introduced to the air compressor speed to assess the robustness of the designed high-potential controller. The disturbance profile of the compressor speed is depicted in Fig. 8(a), with the operating current density maintained at 0.1 A/cm^2 . At 10s, the compressor speed experiences a sudden increase, lasting for 5s, followed by a sudden decrease at 20s, which lasts for 5s. Fig. 8(b) illustrates the control stability of the three developed controllers in the face of disturbances. The T-S LQG controller exhibits minimal robustness since LQG, being a deterministic controller based on the mathematical model, assumes complete knowledge of the system's dynamic characteristics. Consequently, it lacks responsiveness when confronted with unknown disturbances in the system model. In contrast, the T-S MPC controller demonstrates superior robustness. When the system encounters uncertain disturbances, T-S MPC swiftly recovers stability within 2s. This is attributed to MPC compensating for uncertainties during the online optimization process by utilizing real-time measurement data, enabling it to better adapt to the actual dynamics of the system.

4.5. Dynamic response of oxygen partial pressure

To a certain extent, the implementation of cathode recirculation at small current density can have two mutually exclusive effects on the output voltage. An increase in humidity increases the output voltage, while a decrease in oxygen partial pressure decreases the output voltage. However, from the results in Fig. 4, it can be proved that the oxygen partial pressure has a dominant contribution to the reduction of voltage. Fig. 8 detailed shows the dynamic response of the internal oxygen partial pressure during the high potential suppression control. It can be seen that three controllers vary the output voltage by decreasing the oxygen partial pressure compared to no cathode circulation mode. The oxygen partial pressure dynamics shown in Fig. 9(a) can explain some of the phenomena in Fig. 7(b). Specifically, at each current step, the controller regulates the speed of the circulation pump to achieve the target partial pressure of oxygen and thus achieves accurate voltage control. As a result, there is a significant deviation in the voltage during the period from the beginning of the change in oxygen partial pressure to its stabilization. T-S MPC method show the best control effect with the fastest oxygen partial pressure response and no overshoot for each load change. In contrast, PI has significant overshooting and T-S LQG has the slowest response speed. Fig. 9(b) shows the amplified view of the oxygen partial pressure dynamic during loading condition in order to more clearly represent the above results.

4.6. Quantification of control costs and effects

To quantitatively evaluate the control output performance, also known as voltage control level as illustrated in Fig. 8, two error integral criteria are recorded as the integral of squared error (ISE) and the integral of time multiplied by squared error (ITSE), respectively, calculated as shown in equation 36 and 37. For ISE, which reflects the accumulated control error of the output in a finite time t . The ITSE, on the other hand, is more concerned with the error in the late transient response of the controller.

$$ISE = \int_0^t |V_{ctrl}(t) - V_{ref}(t)|^2 dt \quad (36)$$

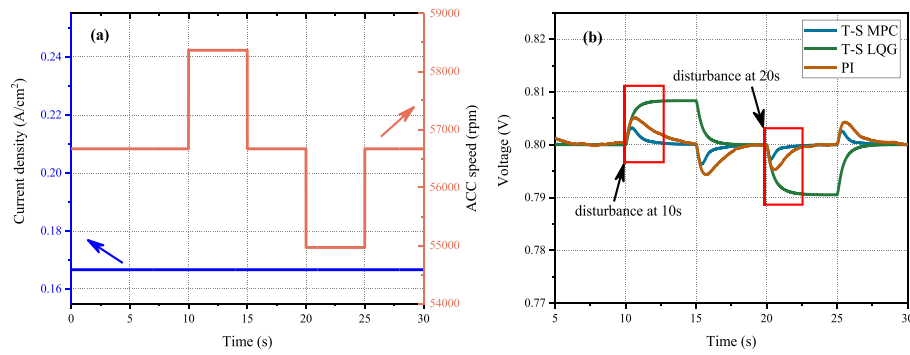


Fig. 8. Robustness analysis of different controllers. (a) Current and disturbance conditions, (b) Voltage control stability demonstration.

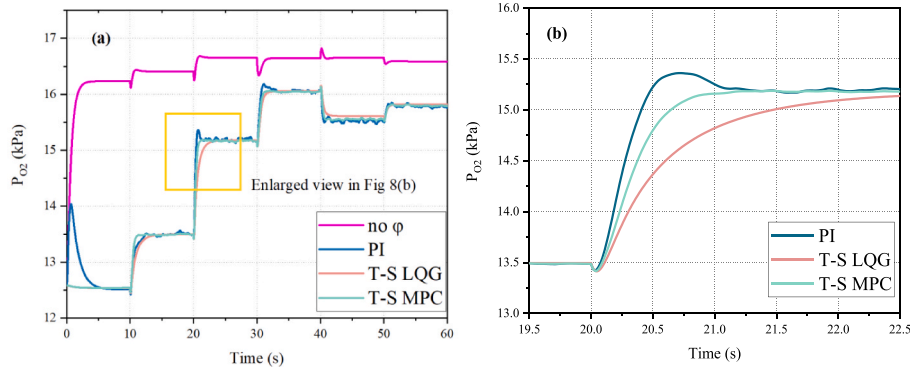


Fig. 9. Oxygen partial pressure response of different control methods. (a) is the oxygen partial pressure dynamic, (b) is the enlarged view of yellow box in Fig. 8(a).

Table 6

Evaluation of the output performance of different controllers.

Criteria	PI	T-S LQG	T-S MPC
ISE	4.2×10^{-4}	2.3×10^{-4}	1.6×10^{-4}
ITSE	3.1×10^{-3}	4.9×10^{-3}	2.8×10^{-3}
TV	4.5×10^{-5}	2.4×10^{-5}	2.0×10^{-5}

scenarios subjected to external disturbances. The control costs and control performance are summarized in Table 6. Compared to the PI and T-S LQG, the control cost TV of the T-S MPC is reduced by 55.5% and 16.7%, respectively. On the contrary, the ISE of the T-S MPC is reduced by 62% and 30.4%, respectively, which represents smaller cumulative errors. The ITSE is reduced by 9.7% and 42.8%, respectively, which represents faster response time.

4.7. Concise comparison

This paper proposes an optimal control scheme for mitigating carbon corrosion and enhancing durability in automotive fuel cells under high potential, utilizing the cathode recirculation method. Up to now, most research has focused on the open-loop control and performance analysis of cathode recirculation, with limited exploration of its closed-loop control performance. To further clarify the effectiveness and superiority of the research results in this paper, we have compared the results with representative published studies. The detailed comparison results are presented in Table 7. Specifically, in Ref. [46], researchers proposed a feedforward + fuzzy PID (SFF + FSPID) algorithm to suppress high potential. Under similar current step amplitudes, it can be observed that there is no difference in static error between the two algorithms;

Table 7

Comparison of control effects between this study and previous research.

Stage	Controllers	Evaluation indicators		
		Overshoots	Static error	Tuning time
Step-up	SFF + FSPID (ref. [46])	0.5 mV	0	2s
	T-S MPC (this study)	0	0	1s
Step-down	SFF + FSPID	2 mV	0	0.5s
	T-S MPC	0	0	1s

both can accurately achieve steady-state tracking. However, in terms of dynamic response, the control algorithm in this paper exhibits almost no overshoot and can rapidly achieve target value tracking. This further confirms the superiority of the approach proposed in this paper for automotive fuel cell systems. Overall, this paper enriches the previous research findings, providing a new solution for the effective control of cathode recirculation in automotive PEMFCs.

5. Conclusion

In this study, the effect of cathode cycling on the voltage characteristics of a proton exchange membrane fuel cell is investigated, and the electrochemical impedance spectroscopy is used to gain insight into the cause from the perspective of internal polarization. On this theoretical basis, in order to reduce the risk of high voltage suffered by automotive fuel cells at low loads and thus further improve the durability, an optimal high-potential controller combining the T-S fuzzy and model predictive control algorithms is developed. The main findings are as follows:

- (1) The relative humidity at the cathode inlet gradually increases with the increase of the cycling ratio, but the rate of humidity increase gradually decreases with the increase of the ϕ . Voltage is more sensitive to changes in the ϕ at high currents than at low currents. At $\phi = 1$, the voltage drop rates are 2.2%, 3%, 4.6%, 5.8% and 7% when the current density is varied from 0 to 0.2A/cm², respectively.
- (2) The electrochemical impedance spectroscopy results show that the increase of the cathode cycling ratio exacerbates the polarization impedance at the mid- and low-frequency. The obstruction of oxygen reduction reaction and the weakening of oxygen diffusion rate are the direct causes of the increase in polarization impedance. On the contrary, the ohmic resistance gradually decreased with the increase of the ϕ , and the ohmic internal resistance decreased by 28.1% when $\phi = 1$.
- (3) T-S fuzzy-based multi-model scheduling method handles the model mismatch problem in developing linear controllers. Compared to PI and T-S LQG, the optimal T-S MPC method has significant advantages in terms of disturbance suppression, fast response and stable tracking. The T-S MPC method reduced the cumulative error by 62% and 30.4% and the dynamic error by 9.7% and 42.8%, respectively, over the previous two methods, while the control costs are reduced by 55.5% and 16.7%, respectively.

Building upon this study, future research can further explore the following aspects:

- (1) Validate the control algorithms by extending the testing from a software platform to an actual hardware platform, evaluating the responsiveness, stability, and accuracy of the algorithms in real-world scenarios.
- (2) Conduct further testing to assess the practical operational effects of cathode recirculation in high-power automotive fuel cell systems.
- (3) Verify the actual impact of cathode recirculation on durability enhancement and assess its real-world effectiveness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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